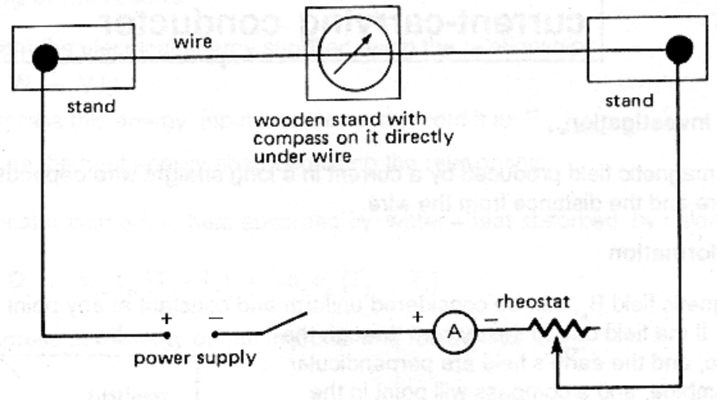
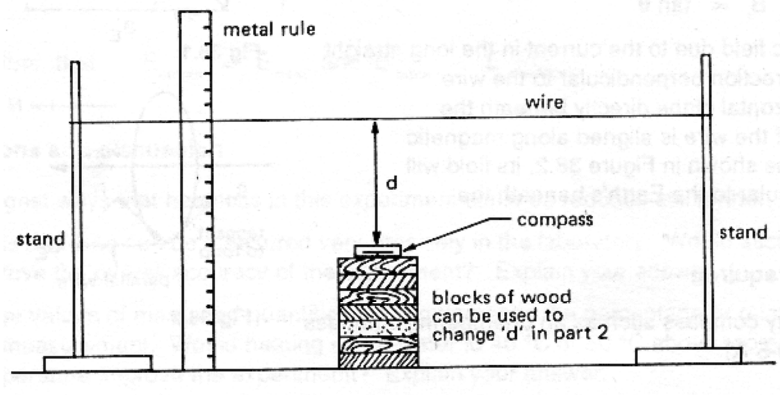
INVESTIGATION: STRENGTH OF EARTH'S MAGNETIC FIELD

AIM To determine the magnetic field strength of the Earth.

METHOD

1. Set up the following circuit with the thick copper wire aligned N - S in the Earth's field.





2. Arrange the wire at a set distance above the compass. Record this distance.
3. Place a 1.00 A current through the wire and record the deflection (θ_1).
4. Reverse the direction iof the current and record θ_2.
5. Repeat for several other currents up to 5.00 A.
6. Graph tan θ against I and use it to determine the Earth's magnetic field strength.

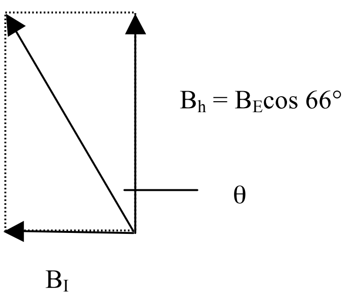
BACKGROUND

The magnetic field of a wire is given by:

$$B_I = \frac{\mu_o}{2\pi}\frac{I}{d}$$

where $\mu o = 4\pi \times 10^{-7}\ NA^{-2}$.

This field is at right angles to the wire, and will be at right angles to the horizontal component of the Earth's field.



The relationship between B_I and B_E becomes the following.

$$\tan\theta = \frac{B_I}{B_E} = \frac{\mu_o I}{2\pi d}\frac{1}{B_E\cos 66°}$$

RESULTS

I (A)	**θ_1**	**θ_2**	**Average θ**	**tan θ**

DISCUSSION Consider the following points.

- Sources of error.
- Accuracy compared to expected result.

CONCLUSION Write a suitable conclusion.